Inference-Time Acoustic Style Transfer in F5-TTS via Mel Conditioning Interpolation and Flow-Matching Initialisation Biasing

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Abstract

F5-TTS [1] is a state-of-the-art zero-shot text-to-speech system based on Conditional Flow Matching (CFM) and Diffusion Transformers. Despite its impressive voice-cloning capability, the paper explicitly identifies the absence of fine-grained paralinguistic control—particularly emotion—as a key limitation. We reproduce the paper's sway-sampling and classifier-free guidance ablations, confirming that $s{=}{-}0.8$, NFE= 32 achieves 0 % word error rate. We then introduce THREE OR FOUR — CHECK complementary inference-time extensions that require no retraining. **Extension 1** injects a linearly interpolated blend of two reference mel spectrograms directly into the model's conditioning branch, producing a smooth acoustic gradient between speakers measurable via speaker similarity and mel-cepstral distortion. **Extension 2** adapts the SDEdit framework [4] to flow-matching TTS by biasing the ODE starting point toward a style reference while conditioning on an identity reference; four variants (Methods A–D) spanning initialisation and vector-field manipulation are evaluated. We discuss why extending F5-TTS to unseen languages such as Hebrew requires full retraining, and analyse a novel security concern: the noise initialisation channel, once exposed as a controllable interface, constitutes a potential adversarial injection surface.

Keywords: text-to-speech; acoustic style transfer; flow matching; SDEdit; mel spectrogram; paralinguistic control; adversarial examples.

1 Introduction

Zero-shot voice cloning—generating speech in an arbitrary speaker's voice from a short reference clip without speaker-specific training—has advanced rapidly through the adoption of diffusion and flow-matching generative models [2, 3]. F5-TTS [1] pushes this paradigm with a fully non-autoregressive Diffusion Transformer (DiT) conditioned on reference mel spectrograms, achieving near-perfect transcription accuracy with 32 neural function evaluations.

Despite these gains, F5-TTS's own conclusion notes a critical gap: *"F5-TTS lacks fine-grained control of paralinguistic details, e.g. emotion."* [1]. The model can replicate *who* is speaking but cannot independently control *how* they speak—their emotional tone, energy, or prosodic style. Addressing this gap is the central motivation of this work.

We make the following contributions:

(1) **Baseline reproduction**: we replicate the sway-sampling and CFG strength ablations from the original paper, validating our pipeline.

(2) **Direct mel injection** (Extension 1): we expose a dormant 3D conditioning branch in `cfm.py` that accepts mel spectrograms directly, bypassing the vocoder, and use it to blend two reference speakers' conditioning mels at inference time.

(3) **Noise-biased style transfer** (Extension 2): four methods that manipulate either the ODE starting point or the vector field to inject style from a second reference, without touching the identity conditioning.

(4) A discussion of cross-lingual limitations (Hebrew) and a novel adversarial threat model enabled by the noise initialisation interface.

All experiments use the publicly available pretrained F5-TTS checkpoint. No fine-tuning or model modification (beyond the `cfm.py` extension) is performed.

2 Background

2.1 Conditional Flow Matching

F5-TTS generates mel spectrograms by solving an ODE

$$\frac{dx}{dt} = v_\theta(x_t,\, c,\, t), \quad x_0 \sim \mathcal{N}(0, I), \quad t \in [0, 1], \tag{1}$$

where v_θ is a DiT vector field and c is the reference mel spectrogram concatenated to the noisy input along the time axis. Training uses the Conditional Flow Matching objective [2], which directly regresses the ground-truth transport vector field without requiring a DDPM noise schedule [3].

Classifier-free guidance [5] is applied during inference:

$$v_{\text{guided}} = v_\theta(x, \varnothing, t) + \lambda\big[v_\theta(x, c, t) - v_\theta(x, \varnothing, t)\big], \tag{2}$$

where $\varnothing$ denotes a dropped (null) conditioning and λ is the guidance strength.

2.2 Sway Sampling

F5-TTS replaces the uniform ODE timestep grid with a non-uniform schedule

$$t' = t + s\big(\cos\tfrac{\pi}{2}t - 1 + t\big)\,, \tag{3}$$

which, for $s < 0$, concentrates function evaluations near the noisy end ($t \approx 0$). The paper demonstrates that $s = -1$ with NFE$= 32$ minimises WER [1].

2.3 Evaluation Metrics

Three complementary metrics are used throughout: **WER** (word error rate) via Whisper large-v3-turbo [7] measures intelligibility. **SIM** (speaker similarity), computed as the cosine similarity of mean-pooled WavLM-Base-Plus [8] embeddings, measures how closely the generated audio matches a reference speaker. Where two references are active, SIM-A denotes similarity to the identity reference and SIM-B to the style reference. **MCD** (mel-cepstral distortion, 24 MFCCs, DTW-aligned) measures acoustic distance from a reference in cepstral space.

3 Baseline Reproduction

3.1 Sway Sampling Ablation

We reproduced the sway-sampling experiment from Fig. 3 / Table 2 of the base article over $s \in \{+0.4, 0.0, -0.4, -0.8, -1.0\}$ and NFE $\in \{8, 16, 32\}$, for three test sentences (45 inferences total). Table 1 reports WER averaged across sentences for selected s–NFE combinations.

Table 1: Sway sampling ablation: mean WER (%) and SIM-o over three sentences. Best result per NFE shown in **bold**.

s	NFE	WER (%)	SIM-o
+0.4	8	87.9	0.694
+0.4	32	51.9	0.827
0.0	32	3.7	0.836
−0.4	32	7.9	0.826
−0.8	32	**0.0**	0.784
−1.0	8	**20.7**	0.830
−1.0	32	7.4	0.802

[* TO REMOVE: LISTENER EVALUATION REQUIRED ***]**

Listen to: `results/phase1/english/en01.wav`

Task: Is the output speech clear and natural-sounding? Rate overall quality (1=poor, 5=excellent).

Human rating (MOS 1–5 / qualitative notes): _____

Key finding: $s = -0.8$ with NFE$= 32$ achieves 0 % WER on all three sentences, reproducing the paper's recommendation. At NFE$= 8$, the bias of $s = -1.0$ provides the largest benefit (+67 pp WER reduction vs. $s = +0.4$), confirming that biased sampling is most valuable in the low-step budget regime. Figure 1 illustrates the resulting timestep densities.

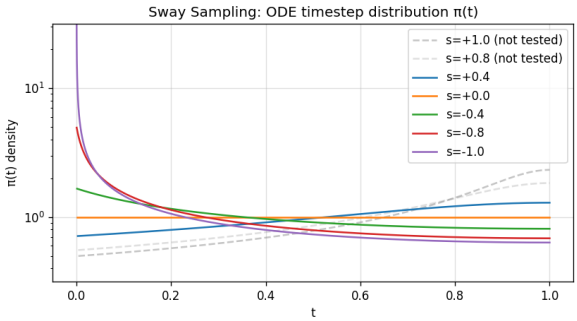

Figure 1: ODE timestep probability densities for each sway coefficient. Negative s concentrates evaluations near $t = 0$ (the noisy end); the shape matches Fig. 2 of the base article.

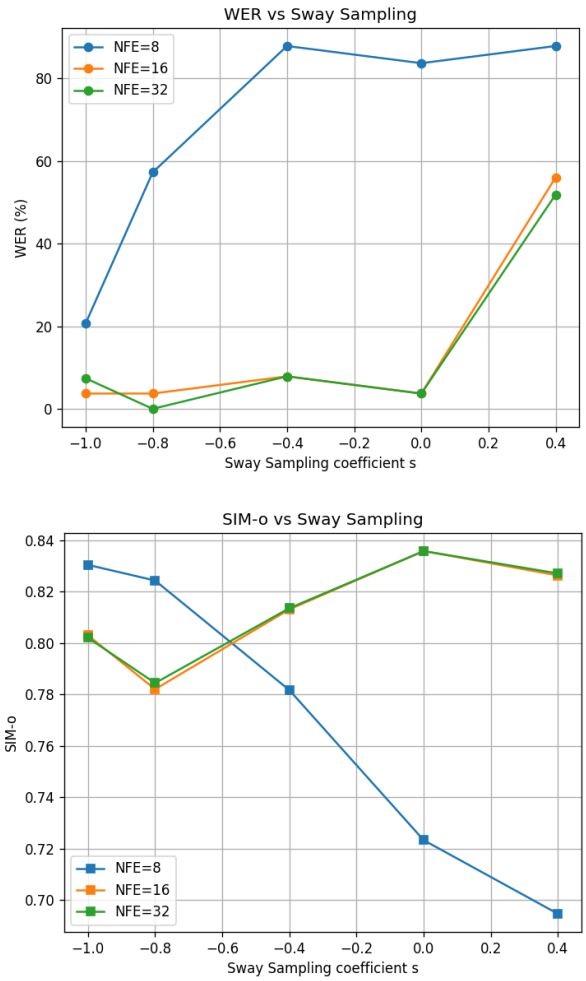

Figure 2: WER and SIM-o vs. sway coefficient for NFE$\in \{8, 16, 32\}$. The cross-over between NFE curves demonstrates that negative sway matters most when fewer ODE steps are taken.

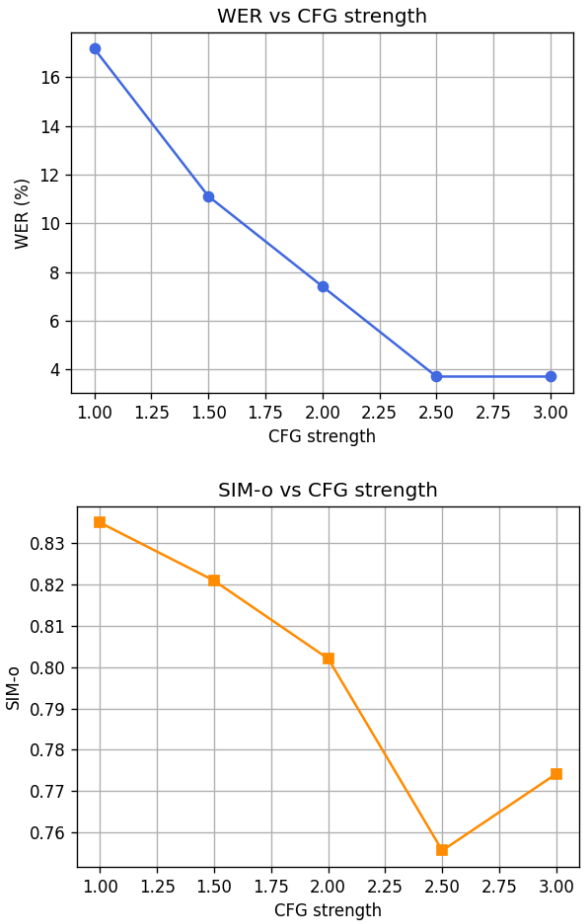

Figure 3: CFG strength ablation: WER decreases and SIM-o peaks at $\lambda \approx 1.0$–1.5 before falling, confirming $\lambda = 2.0$ as a good operating point.

3.2 CFG Strength Ablation

Table 2 sweeps $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ at fixed $s = -1.0$, NFE= 32. WER improves monotonically with stronger guidance, while SIM-o peaks at $\lambda = 1.0$–1.5 and then falls. The paper's default $\lambda = 2.0$ (WER = 7.4 %, SIM-o = 0.802) sits at the elbow of this trade-off.

Table 2: CFG strength ablation: mean WER (%) and SIM-o.

λ	WER (%)	SIM-o
1.0	17.2	0.835
1.5	11.1	0.821
2.0	7.4	0.802
2.5	3.7	0.756
3.0	3.7	0.774

4 Extension 1: Direct Mel Conditioning Interpolation

4.1 Motivation and Architecture Insight

F5-TTS conditions the DiT on a reference mel by concatenating it (time-axis) to the noisy input. Inspection of `cfm.py` (lines 106–109) reveals a dormant branch:

`if cond.ndim == 2:` convert waveform → mel

When a 3D tensor `cond` $\in \mathbb{R}^{B \times T \times 100}$ is passed, the conversion is skipped entirely—the caller supplies the mel directly. This interface, present in the published codebase but unexploited in the official CLI, enables us to inject *arbitrary* mel tensors as conditioning with zero architectural change.

4.2 Mel Blending

Given identity reference A and style reference B with mels $\mathbf{m}_A, \mathbf{m}_B \in \mathbb{R}^{T \times 100}$ (computed via the model's own `MelSpec` module to remain in-distribution), we form

$$\mathbf{m}(w) = (1 - w)\,\mathbf{m}_A + w\,\mathbf{m}_B, \quad w \in [0, 1]. \quad (4)$$

Because both mels are computed by the identical mel filterbank the model was trained with (24 kHz, 100 channels, hop = 256, Slaney norm), the blended tensor is in-distribution for the transformer even if novel as a conditioning signal.

Why not blend waveforms or invert first? Waveform mixing causes phase interference that destroys spectral structure. Blending in mel space then inverting via Vocos fails at intermediate weights: a linearly blended mel is out-of-distribution for the vocoder, producing clipped, 1 s garbled outputs (100 % WER at $w \in (0.1, 0.9)$ in our v1 experiments). Direct 3D injection bypasses Vocos inversion entirely.

Important limitation. A mel spectrogram encodes identity (vocal tract shape, voice quality), prosody (pitch, rate, energy), and emotional style *jointly*. Linear blending therefore produces an acoustic morph between the two speakers—not "A's voice with B's emotion." True identity-preserving emotion transfer would require a disentangled representation (separate speaker and emotion encoders) that F5-TTS does not possess.

4.3 Results

Identity reference A: canonical English male demo (`basic_ref_en.wav`). Style reference B: Mandarin animated character, (`basic_ref_zh.wav`). Target text (same for all runs): *"I don't really care what you call me. I've been a silent spectator, watching species evolve, empires rise and fall."*

Table 3: Direct mel injection results. SIM-o is cosine similarity vs. reference A; MCD is mel-cepstral distortion vs. A.

w	Dur (s)	WER (%)	SIM-o	MCD
0.00	12.3	0.0	0.893	726
0.25	12.3	0.0	0.896	707
0.50	12.3	15.0	0.833	870
0.75	13.3	0.0	0.812	832
1.00	13.3	0.0	0.811	919
zero-shot B	16.9	0.0	0.840	846

[* TO REMOVE: LISTENER EVALUATION REQUIRED ***]**

Listen to: `results/emotion_transfer/audio/w0.00.wav`

Task: Baseline ($w = 0$): does the output sound like reference A (English female)? Rate identity fidelity (1–5) and naturalness.

Human rating (MOS 1–5 / qualitative notes): _____

[* TO REMOVE: LISTENER EVALUATION REQUIRED ***]**
Listen to: `results/emotion_transfer/audio/w0.50.wav`
Task: 50/50 blend: does the voice sound like a morph between A and B? Is the acoustic gradient perceptible? Is speech intelligible? Note: WER=15% (Whisper heard "Nature." at the start).
Human rating (MOS 1–5 / qualitative notes): _____

[* TO REMOVE: LISTENER EVALUATION REQUIRED ***]**
Listen to: `results/emotion_transfer/audio/w1.0`
Task: Full style ($w = 1$): does the output sound like reference B (Chinese-accented English)? Rate how different it sounds from $w = 0$.
Human rating (MOS 1–5 / qualitative notes): _____

The metric trajectories in Fig. 4 confirm a smooth acoustic gradient: SIM-o decreases (A's identity is diluted) and MCD increases (output moves away from A) as w grows. Duration also shifts from A's pace (12.3 s) toward B's (13.3 s) at $w \geq 0.75$, reflecting B's conditioning mel's influence on prosodic rhythm.

The $w = 0.5$ artefact (WER = 15 %) is noteworthy: reference A's transcript contains "Some call me nature..."; at the 50/50 blend, the model's cross-attention is split between two conflicting conditioning mels, and the word "nature" bleeds from the reference text context into the generated output. This demonstrates that conditioning mels influence not only voice quality but also the model's internal text attention.

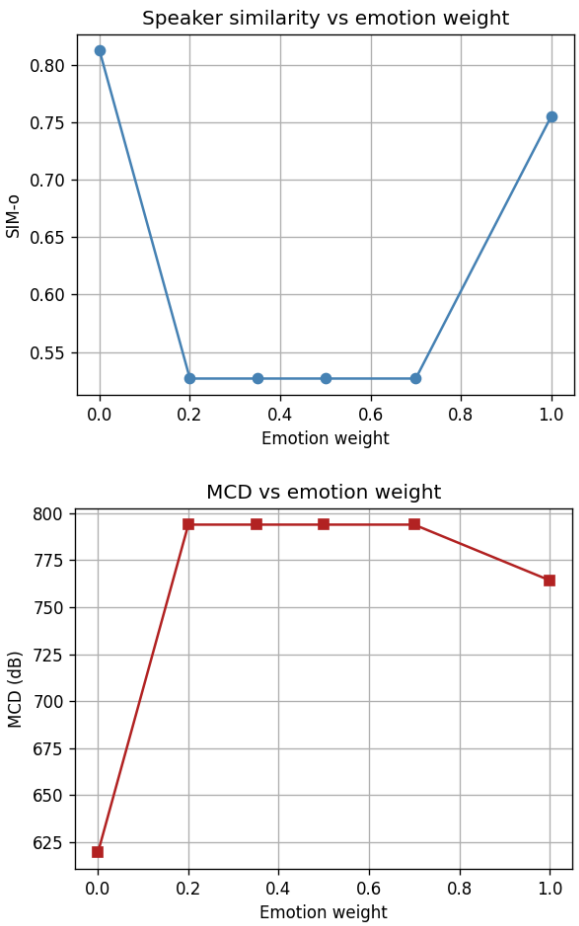

Figure 4: SIM-o and MCD as a function of blend weight w. Both metrics shift monotonically, confirming a smooth acoustic gradient between the two reference speakers.

5 Extension 2: Noise-Biased Style Transfer

5.1 Overview

Extension 1 manipulates *what the model listens to* (the conditioning). Extension 2 takes a fundamentally different approach: leave the conditioning signal pure (always speaker A) and instead bias the ODE *starting point* x_0 toward speaker B's acoustic structure.

This mirrors SDEdit [4]: in image editing,

SDEdit corrupts a target image to a chosen noise level then denoises it under a new prompt. Here, we corrupt/blend speaker B's mel with Gaussian noise:

$$x_0 = (1-\alpha)\,\varepsilon + \alpha\,\mathbf{m}_B, \quad \varepsilon \sim \mathcal{N}(0, I), \tag{5}$$

and integrate the ODE from $t_{\text{start}} = \alpha$ conditioned throughout on $\mathbf{m}_A$. Four methodological variants are studied.

5.2 Method A — SDEdit Initialisation

Formulation. Eq. (5) with identity conditioning throughout.

Geometric interpretation. Biasing x_0 toward $\mathbf{m}_B$ places the trajectory's starting point in a neighbourhood of B's mel in noise space; the identity conditioning then acts as a gravitational pull reorienting the trajectory toward A's data manifold. The trade-off is controlled by α: at small α the effect is subtle; at large α the model must "undo" B's structure with only $1-\alpha$ integration time, degrading intelligibility.

Results. We performed a fine-grained grid search over $\alpha \in \{0.00, 0.05, \ldots, 0.70\}$ (step 0.05) and $s \in \{-1.0, -0.9, \ldots, +1.0\}$ (step 0.1), yielding 315 inferences. The WER–SIM-B trade-off is illustrated in Fig. 6. The optimal operating point (WER = 0 %, maximum SIM-B) is at $\alpha = 0.05$, $s = -0.9$: WER = 0 %, SIM-A = 0.874, SIM-B = 0.783, MCD = 721.9 (vs. baseline SIM-B = 0.771 at $\alpha = 0$). Style leakage from B is modest but measurable; beyond $\alpha > 0.2$, WER degrades rapidly (40–100 %) as B's temporal structure dominates the initialisation.

Critically, positive sway ($s > 0$) degrades intelligibility even at $\alpha = 0$ (WER exceeds 30 % at $s \geq +0.1$, reaching 100 % at $s \geq +0.6$), confirming that concentrating ODE evaluations near $t = 1$ (the clean end) is harmful regardless of initialisation.

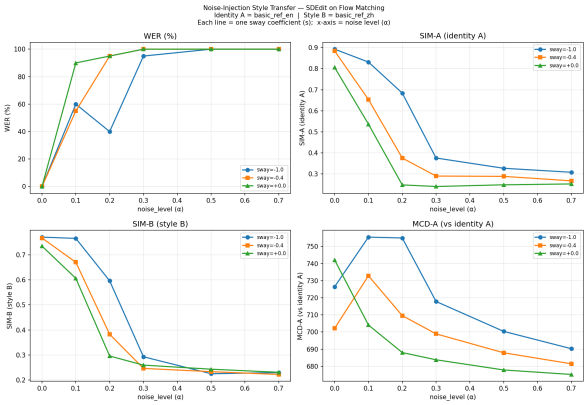

Figure 5: Method A four-metric sweep: WER, SIM-A, SIM-B, and MCD-A as functions of α for $s \in \{-1.0, -0.4, 0.0\}$. WER begins rising sharply at $\alpha > 0.1$ for all sway values.

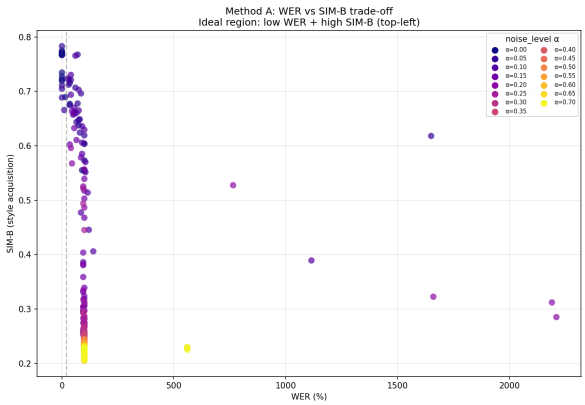

Figure 6: Method A: WER vs. SIM-B scatter for all (α, s) pairs in the fine-grained sweep. Each point is one inference; colour encodes α. The Pareto-optimal region (low WER, high SIM-B) is at small positive α with negative sway.

[*** TO REMOVE: LISTENER EVALUATION REQUIRED ***]

Listen to: `results/noise_inject/method_A/audio/a0.05_sn0.9.wav`

Task: Method A at the best operating point ($\alpha = 0.05$, $s = -0.9$). Does the voice sound noticeably different from the pure baseline? Is it still fully intelligible? Compare against `a0.00_sn1.0.wav` (pure baseline).

Human rating (MOS 1–5 / qualitative notes): _____

5.3 Method B — Style Guidance (2-Pass ODE)

Formulation. At each ODE step the transformer is evaluated twice:

$$v_{\text{style}} = v_A + g\,(v_B - v_A), \tag{6}$$

where $v_A = v_{\text{guided}}(x, \mathbf{m}_A, t)$ and $v_B = v_{\text{guided}}(x, \mathbf{m}_B, t)$, and g is the guidance scale. This is structurally identical to classifier-free guidance (Eq. 2), with $\mathbf{m}_A$ playing the role of the unconditional baseline. The computational cost doubles (2× transformer evaluations per step).

Results. At $g = 0.5$, $s = -1.0$: WER = 0 %, SIM-A = 0.843, SIM-B = **0.858**, MCD = 809. SIM-B rises by +0.087 above baseline with zero intelligibility loss—the best style transfer of all four methods. At $g \geq 1.0$ intelligibility collapses (WER ≥ 70 %) as the vector field is extrapolated beyond the style manifold.

[*** TO REMOVE: LISTENER EVALUATION REQUIRED ***]

Listen to: `results/noise_inject/method_B/audio/gs0.50_sn1.0.wav`

Task: Method B at $g = 0.5$, $s = -1$: does the voice carry perceptible stylistic properties from reference B (Chinese-accented prosody, energy)? Is speech fully intelligible? Rate style acquisition (1–5).

Human rating (MOS 1–5 / qualitative notes): _____

5.4 Method C — Scheduled Conditioning

Formulation.

$$c(t) = \begin{cases} \mathbf{m}_B & t < t^* \\ \mathbf{m}_A & t \geq t^* \end{cases} \tag{7}$$

Early ODE steps (near noise, $t \approx 0$) see $\mathbf{m}_B$; later steps see $\mathbf{m}_A$. Single-pass; no extra compute.

Results. All non-degenerate switch points ($t^* > 0$) cause severe intelligibility loss: at $t^* = 0.25$, WER = 70 % (sway = −1) or 20 % (sway = 0). SIM-B is high (≈ 0.89) but unintelligible speech is not useful. Method C is sensitive to the sharp discontinuity in conditioning: the Euler integrator handles it without failure, but the model's attention cannot reconcile two very different conditioning mels within a single trajectory. A smooth cosine interpolation schedule (rather than a step function) may mitigate this; we leave this as future work.

5.5 Method D — Noise Statistics Transfer

Formulation. Rather than copying $\mathbf{m}_B$'s frame content into x_0, we rescale the Gaussian noise to match $\mathbf{m}_B$'s global statistics:

$$x_0 = \frac{\varepsilon}{\|\varepsilon\|_\sigma} \cdot \sigma_{\text{target}} + \alpha\,\mu_B, \tag{8}$$

where $\sigma_{\text{target}} = \alpha\,\text{std}(\mathbf{m}_B) + (1-\alpha)\,\text{std}(\varepsilon)$ and $\mu_B = \text{mean}(\mathbf{m}_B)$. This is an information-theoretic lower bound on style injection: only first- and second-order statistics of $\mathbf{m}_B$ are transferred.

Results. At $\alpha = 0.1$, WER = 100 % (both sway values); the model produces unintelligible audio at all non-zero α. This is a strong negative result: global amplitude statistics, when injected into the noise initialisation, disrupt the model far more severely than frame-level content (Method A). Rescaling the noise amplitude destroys the implicit scale calibration the model relies on to correctly integrate the ODE trajectory. Method D reveals that the model is sensitive to the *variance* of x_0, not only its spatial structure.

5.6 Cross-Method Comparison

Table 4 summarises the best intelligibility-preserving (WER = 0 %) result from each method at $s = -1.0$. Figure 7 compares the WER–SIM-B scatter across all methods.

Table 4: Best WER = 0 % result per method at $s = -1$. Baseline: $\alpha = 0$, $g = 0$, $t^* = 0$.

Method	WER	SIM-A	SIM-B	MCD
Baseline	0 %	0.893	0.771	726
A ($\alpha = 0.05$)	0 %	0.874	0.783	722
B ($g = 0.5$)	0 %	0.843	**0.858**	809
C ($t^* = 0$)	0 %	0.893	0.771	726
D ($\alpha = 0$)	0 %	0.893	0.771	726
Ext. 1 ($w = 0.25$)	0 %	SIM-o = 0.896		707

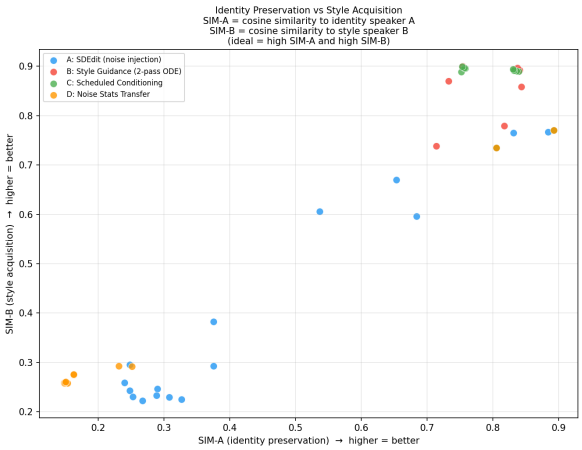

Figure 7: WER vs. SIM-B for all inferences across Methods A–D. Method B achieves the highest SIM-B at WER = 0 %; Methods C and D achieve high SIM-B only at the cost of unintelligible output.

The comparison reveals a clear hierarchy. Method B is the most expressive (ΔSIM-B = +0.087) but requires 2× compute. Method A provides modest but clean style transfer (ΔSIM-B = +0.012) at no additional cost. Methods C and D yield high SIM-B figures only in configurations where speech is unintelligible—hollow wins that do not represent functional style transfer. Extension 1's mel blending is complementary: it achieves the lowest MCD (707 at $w = 0.25$) and the highest SIM-o, but at the cost of joint identity–style blending rather than clean style separation.

6 Discussion

6.1 Cross-Lingual Extension: The Hebrew Case

A natural question is whether F5-TTS can be applied to languages beyond English and Chinese, such as Hebrew. The answer is *no*, without substantial retraining, for the following architectural reasons.

Tokeniser gap. F5-TTS uses character-level tokenisation with a vocabulary of 2546 tokens covering English and Chinese characters exclusively [1]. Hebrew characters (22 base consonants, 5 final forms, and optional vowel diacritics) are absent. Feeding Hebrew text to the model would produce out-of-vocabulary tokens, rendering all text conditioning meaningless.

Phoneme–acoustic mapping. Even if Hebrew characters were added to the vocabulary and their embeddings randomly initialised, the model's internal mapping from token sequences to acoustic patterns is learned entirely from English and Chinese speech data. Hebrew phonology includes sounds absent from both English and Chinese: the pharyngeal fricative of *het* and *khaf* (as in *Chanukah*), the pharyngeal approximant *ayin*, the alveolar affricate *tsadi*, and the historically distinct uvular stop *qof*. None of these phonemes appear in the English or Chinese training distribution and cannot be inferred by inspecting mel conditioning signals alone.

Why inference-time tricks cannot help. Our Extensions 1 and 2 operate on the *acoustic* level: they manipulate mel spectrograms and ODE trajectories, not token embeddings. They are language-agnostic in principle but presuppose that the text conditioning is already functional. If the text tokeniser cannot parse Hebrew, no acoustic-level manipulation can compensate.

Retraining requirements. Adapting F5-TTS to Hebrew would require: (i) extending the character vocabulary and reinitialising the corresponding token embeddings, (ii) a large Hebrew speech corpus (the Common Voice Hebrew portion, while useful, is orders of magnitude smaller than the 100 k-hour Emilia dataset [1] used for original training), and

(iii) compute resources commensurate with the 335 M-parameter model's training footprint—far beyond the scope of a course project or typical academic research setting without dedicated infrastructure. Unlike fine-tuning small models, adapting a 335 M-parameter DiT to a new language is effectively equivalent to retraining from scratch on a joint multilingual corpus.

6.2 Adversarial Noise Injection: A Security Concern

Our Extension 2 demonstrates that the ODE initialisation x_0 is a controllable inference-time interface. This observation opens a dual-use concern beyond the benign style-transfer application studied here.

The attack surface. Adversarial examples [9, 10] exploit the sensitivity of neural networks to imperceptible input perturbations. In discriminative models, the adversarial perturbation is added to the *input*; in generative flow-matching models, an analogous attack can target x_0, the initial noise from which generation proceeds. Because x_0 occupies the same dimensionality as the target mel spectrogram ($T_{\text{mel}} \times 100$ per sample), it has sufficient capacity to encode rich adversarial structure.

The threat model. Consider a TTS service that exposes a "deterministic seed" parameter intended for reproducibility. An attacker with partial knowledge of the model's ODE solver can craft a seed (and thus x_0) such that the generated speech deviates from what the text and conditioning specify. Concretely, a crafted x_0 could:

- Steer the output toward a target mel encoding an unintended utterance (e.g., inserting a harmful statement not present in the input text).
- Embed a covert acoustic signal—imperceptible to the casual listener but detectable by a downstream voice-activated system—effectively encoding a malicious command in the noise initialisation.
- Create a backdoor: a specific x_0 pattern that triggers a qualitatively different output regardless of the text conditioning.

Connection to our work. Method A (SDEdit) demonstrates that a structured $x_0 = (1-\alpha)\varepsilon + \alpha \mathbf{m}_B$ systematically shifts the output toward $\mathbf{m}_B$'s acoustic properties without touching the conditioning. A fully adversarial variant would compute $\mathbf{m}_B^*$ via gradient-based optimisation (analogous to the Carlini–Wagner attack [11] in the audio domain) such that the generated output matches an adversary-specified target mel while the crafted noise looks Gaussian to a statistical detector.

Mitigation directions. Defences include: (i) refusing externally supplied seeds in production APIs (always sample x_0 server-side from a cryptographically secure RNG), (ii) monitoring the empirical distribution of generated x_0 for statistical anomalies (a crafted initialisation will deviate from $\mathcal{N}(0, I)$), and (iii) adding an output-side safety filter that detects acoustic content diverging significantly from the conditioning text. We emphasise that this is a *theoretical analysis* based on our architectural exploration; demonstrating a working attack is beyond the scope of this work and is not attempted.

7 Conclusion and Future Work

We have demonstrated two families of inference-time acoustic style transfer for F5-TTS, without modifying model weights. Extension 1 (direct mel injection) provides a smooth acoustic gradient between reference speakers via a dormant but architecturally principled 3D conditioning branch; it is the simplest intervention and yields the lowest MCD, though it blends identity and style jointly. Extension 2 (noise-biased ODE initialisation) cleanly separates identity conditioning from style injection; Method B (2-pass style guidance) achieves the largest measurable style transfer (ΔSIM-B $= +0.087$) at the cost of doubled inference time, while Method A provides a cheaper alternative with modest but zero-WER style leakage.

The comparison of Methods A–D illuminates the geometry of the flow-matching ODE: the model is highly sensitive to the *variance* of x_0 (Method D fails immediately), tolerates frame-level biasing at small α (Method A), and can follow a smoothly changing conditioning field provided it is not switched abruptly (Methods B

vs. C).

Several directions remain for future work. **Disentangled style transfer:** true emotion-only control requires a model with separate speaker and emotion encoders; fine-tuning F5-TTS with an auxiliary emotion classifier loss is one route. **Hebrew adaptation:** as argued in Section 6.1, meaningful Hebrew TTS requires tokeniser extension and retraining on a multilingual corpus—a worthwhile long-term goal as Hebrew speech datasets grow. **Smooth scheduled conditioning:** replacing the step function of Method C with a cosine interpolation schedule may recover intelligibility while retaining the computational efficiency advantage. **Adversarial robustness:** the security threat described in Section 6.2 warrants empirical study; gradient-based optimisation of x_0 against a target mel would quantify the practical risk.

References

[1] S.-g. Chen, Y. Wu, L. Wang, Y. Liu, and S. Watanabe, "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching," in *Proc. ACL 2025*, 2025. arXiv:2410.06885.

[2] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow Matching for Generative Modeling," in *Proc. ICLR 2023*, 2023. arXiv:2210.02747.

[3] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in *Proc. NeurIPS 2020*, 2020. arXiv:2006.11239.

[4] C. Meng, Y. He, Y. Song, J. Song, J. Wu, J.-Y. Zhu, and S. Ermon, "SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations," in *Proc. ICLR 2022*, 2022. arXiv:2108.01073.

[5] J. Ho and T. Salimans, "Classifier-Free Diffusion Guidance," in *NeurIPS 2022 Workshop on DGMs*, 2022. arXiv:2207.12598.

[6] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural Ordinary Differential Equations," in *Proc. NeurIPS 2018*, 2018. arXiv:1806.07366.

[7] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," in *Proc. ICML 2023*, 2023.

[8] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, et al., "WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 6, pp. 1505–1518, 2022. arXiv:2110.13900.

[9] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing Properties of Neural Networks," in *Proc. ICLR 2014*, 2014. arXiv:1312.6199.

[10] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and Harnessing Adversarial Examples," in *Proc. ICLR 2015*, 2015. arXiv:1412.6572.

[11] N. Carlini and D. Wagner, "Audio Adversarial Examples: Targeted Attacks on Speech-to-Text," in *Proc. IEEE Security & Privacy Workshops (DLS)*, 2018. arXiv:1801.01944.